

Florence Urban Land Use Dashboard

1. Project Overview

This project analyses land-use patterns in Florence using OpenStreetMap data, QGIS, Excel, and Power BI. The objective was to identify the main land-use categories and create visualisations to support urban planning and spatial analysis.

2. Data Source

- OpenStreetMap (OSM)
- Florence land-use data extracted using QuickOSM in QGIS

3. Methodology

1. Extracted land-use data from OpenStreetMap using QuickOSM.
2. Processed and visualised the data in QGIS.
3. Calculated polygon areas by land-use category.
4. Generated statistical summaries in CSV format.
5. Created charts and dashboard visualisations in Excel and Power BI.

4. Key Findings

- Forest was the largest land-use category.
- Residential land represented the largest urban land-use type.
- Orchard and farmland areas occupied a significant proportion of the study area.
- Florence contains substantial green and agricultural spaces alongside urban development.

5. Tools Used

- QGIS
- QuickOSM
- Excel
- Power BI
- OpenStreetMap

6. Skills Demonstrated

- GIS Analysis
- Spatial Data Processing

- Data Cleaning
- Statistical Analysis
- Data Visualisation
- Dashboard Development

7. Project Outputs

- Land-use map created in QGIS
- Land-use statistics table
- Bar chart
- Pie chart
- Power BI dashboard

8. Conclusion

This project demonstrates the ability to collect, process, analyse, and visualise geospatial data using industry-standard GIS and business intelligence tools. The workflow transformed raw spatial data into actionable insights suitable for urban planning and decision-making.



